Model Evaluation Showdown

Machine Learning Assignment 9

October 7, 2026

1 Introduction

In this assignment, we train three distinct classifiers: Logistic Regression, K-Nearest Neighbors (KNN), and a Decision Tree, on the same synthetic binary classification dataset. We evaluate all three models across multiple metrics (Accuracy, Precision, Recall, F1-Score) and examine their confusion matrices to simulate a real-world model selection scenario.

2 Metrics Comparison Table

Below is the comparison table detailing the performance metrics for each classifier on the unseen test set:

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.860	0.894	0.808	0.849
KNN	0.940	0.971	0.904	0.936
Decision Tree	0.880	0.910	0.836	0.871

Table 1: Model Evaluation Metrics.

3 Confusion Matrices

The confusion matrices provide deeper insight into how each model classifies True Positives, True Negatives, False Positives, and False Negatives.

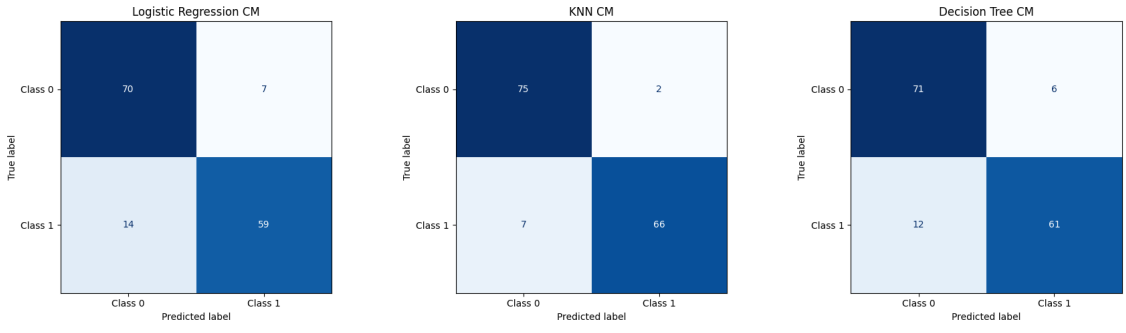


Figure 1: Confusion Matrices for Logistic Regression, KNN, and Decision Tree.

4 Recommendation and Interpretation

Based on the quantitative evaluation metrics and the confusion matrices, I recommend deploying the **KNN (K-Nearest Neighbors)** classifier for this specific dataset.

Justification for KNN

The KNN model strictly dominates the other two models across all primary metrics:

- It boasts the highest overall **Accuracy** (94.0%), meaning it makes the fewest total mistakes.
- It has the highest **Precision** (97.1%), minimizing False Positives (as seen in its confusion matrix, it only predicted 2 False Positives).
- It achieves the highest **Recall** (90.4%), minimizing False Negatives better than the linear or tree-based approach.

When would I choose differently?

While KNN is the clear winner here, there are specific real-world contexts where I would choose one of the other models instead:

- **When latency/deployment speed matters:** KNN is a "lazy learner." To make a single prediction, it must compute distances against the entire training dataset. If we needed to deploy this on a highly-trafficked web server requiring microsecond inference times, I would choose **Logistic Regression**. It simply evaluates a dot product, making it immensely faster at inference time.
- **When interpretability is required:** If this model was used for medical diagnosis, credit lending, or any regulated industry where we must explain *why* a prediction was made to a human user, KNN is a "black box." In this scenario, I would choose the **Decision Tree**. Despite its lower accuracy (88%), a Decision Tree provides transparent, human-readable logic rules (e.g., "If Feature X $>$ 2, then Class 1") that can be audited.